i-ELOOP

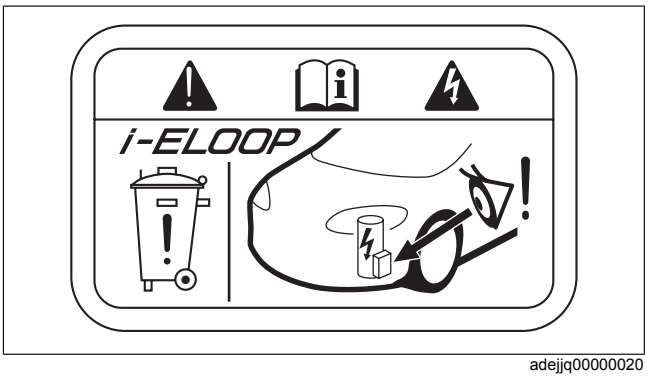
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Purpose of Adoption

- For improved fuel economy, a large amount of electrical power is instantaneously generated using kinetic energy during deceleration and it is retrieved as electrical energy and reused, which reduces the power generation load on the engine.

Service Point

- There are systems which differ from power voltage. For the removal/installation or replacement of **13—25 V** parts (capacitor, DC-DC converter, generator) in particular, always disconnect the negative battery terminal and pull out the capacitor safety plug.
- If the DC-DC converter is replaced (capacitor replaced at same time) with the capacitor voltage decreased, the mode switches to the pre-charge mode for charging the capacitor after connecting the battery or starting the engine.
- After starting the engine when replacing the DC-DC converter, verify that there is no DTC record. If a DTC is stored, always clear the DTC using the M-MDS.
- When the engine is started the first time after replacing the DC-DC converter, the capacitor is changed by generating power at the generator. In addition, if the capacitor voltage is low and the battery voltage is low, it takes several seconds to complete the capacitor charge.
- During this time, the battery voltage may further decrease and each electrical component may store a DTC because electrical load is supported by only the battery.
- Because examination for the capacitor using the tester or i-ELOOP system is not possible, the examination is performed based on the external appearance (capacitor damage, capacitor connector damage or corrosion).
- Store the capacitor in an upright position. However, the part can be tilted for a short period during installation.
- If there is external appearance damage or deformation because the capacitor was dropped, replace it with a new one (no external appearance damage and deformation).
- The service parts are delivered with the short wire installed to prevent static discharge. Remove the short wire before installing to the vehicle.
- The following label is attached on the upper part of the bumper. For vehicles with the capacitor, discharge operation is performed using a capacitor discharge resistance box when the capacitor is discarded. Discard the same way as other parts after discharge.



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Common Inquiry

- Not applicable

Other

- Size reduction of battery
  - — A DC-DC converter and capacitor with improved performance have been adopted, reducing the size of the lead battery.
- Placement of DC-DC converter in engine compartment
  - — The size reduction of the wiring harness connected to the i-ELOOP related-parts has been realized by reducing the size of the lead battery and placing the DC-DC converter in the available space.
- Changing pre-charge method
  - — If the engine is started with the capacitor voltage decreased after vehicle is left for long periods, the capacitor is shut off and the mode is switched to bypass mode. Due to this, even if the capacitor voltage is low, the power supply can be assured immediately.
  - — The pre-charge operation is performed via the charge circuit in the DC-DC converter with the capacitor and generator shut off electronically.